# 100 Days of Python

Day 56 — Practice Assignment: Debugging

*Debugging routine: reproduce it, read the error, locate the line, change one thing, run it again.*

**1. This program has two syntax errors. Run it, read what Python says, and fix both.**

```
def greet(name)
    print("Hello, " + name
greet("Asha")
```

*Hint: look at the end of the def line, and count the brackets on the print line.*

**2. Read this traceback and answer four things: the error type, the function it happened in, the line number, and what in the input caused it.**

```
Traceback (most recent call last):
  File "main.py", line 7, in <module>
    print(average_pass([20, 30, 10]))
  File "main.py", line 3, in average_pass
    return sum(passed) / len(passed)
ZeroDivisionError: division by zero
```

*Hint: read the last line first, then the line just above it.*

**3. This should print 3, because 45, 40 and 90 are all passes (40 or above). It prints 2. Add a print line inside the loop to watch what happens, find the bug, and fix it.**

```
def count_passed(marks):
    count = 0
    for m in marks:
        if m > 40:
            count += 1
    return count
print(count_passed([45, 40, 34, 90]))
```

*Hint: no error is shown, so this is a logic error. Which mark is being treated wrongly?*

**4. This program prints all three numbers and then crashes. Name the error, explain why it happens after the correct output, and fix it.**

```
nums = [10, 20, 30]
for i in range(len(nums) + 1):
    print(nums[i])
```

*Hint: how many positions does nums have, and what is the last value i reaches?*

**5. Run this program and read the error. Explain what NoneType means here, and fix the function so that it prints 6.**

```
def add(a, b):
    a + b
result = add(2, 3)
print(result + 1)
```

*Hint: this is the Day 47 mistake. What does a function give back when it has no return?*